Few-Shot Learning-Based Plant Leaf Disease Classification Model Development and Performance Evaluation

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Abstract & Introduction

Plant diseases are one of the leading causes of reduced agricultural productivity worldwide, threatening food security for millions. Traditional detection methods are often slow, expensive, and heavily dependent on expert knowledge, making them impractical for small-scale farmers, especially in developing regions.

To address this, we propose a **lightweight few-shot learning framework** combining a **Prototypical Network** with a custom **SqueezeNet** backbone of only **73,032 parameters**. our model learns from just a handful of examples, achieving up to **95.63% accuracy** across multiple plant species.

Dataset Samples

Plant	Disease Categories (Number of Images)	Total
Mango	**Anthracnose (500), Bacterial canker (500), Cutting weevil (500), Die back (500), Gall midge (500), Powdery mildew (500), Sooty mould (500), Healthy (500)**	**4000**
Rice	**Bacterial leaf blight (40), Brown spot (40), Leaf smut (40)**	**120**
Potato	**Early blight (1939), Late blight (1939), Healthy (1824)**	**5702**
Apple	**Apple scab (2016), Black rot (1987), Cedar apple rust (1760), Healthy (2008)**	**7771**
Pepper Bell	**Bacterial spot (1913), Healthy (1988)**	**3901**

Figure 1. Sample images from the plant leaf dataset (Potato, Mango, Pepper Bell, Apple, Rice).

Methodology

Core Framework: The proposed system uses a compact **SqueezeNet** backbone (~73K parameters) integrated with a **Prototypical Network (ProtoNet)**. This enables accurate plant leaf disease classification from only a few labeled examples per class ideal for real-world deployment where labeled data is scarce.

Feature Extraction: The SqueezeNet backbone extracts compact yet discriminative features from input leaf images. Its **Fire Module** architecture (shown below) significantly reduces the parameter count while preserving strong representational capacity compared to larger networks such as VGG or ResNet.

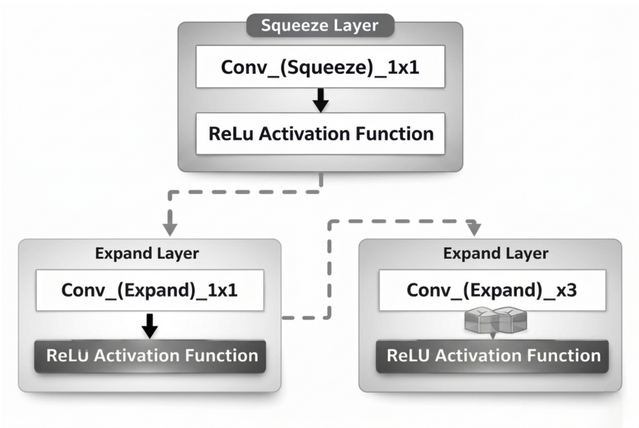

Figure 2. SqueezeNet Fire Module Architecture.

Attention Mechanism: Channel Attention selectively emphasizes the most informative feature channels, while **Spatial Attention** focuses on the exact regions where disease symptoms appear such as spots, lesions, or discoloration.

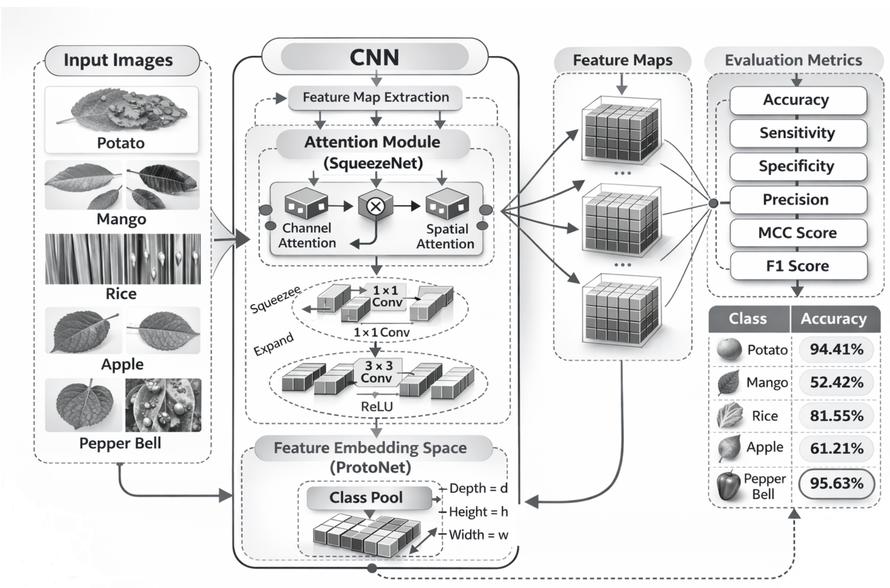

Figure 3. Proposed SqueezeNet-based Prototypical Network Architecture.

Results & Analysis

Our model achieves strong diagnostic performance with limited labeled data, outperforming several existing approaches while maintaining an ultra-compact model size.

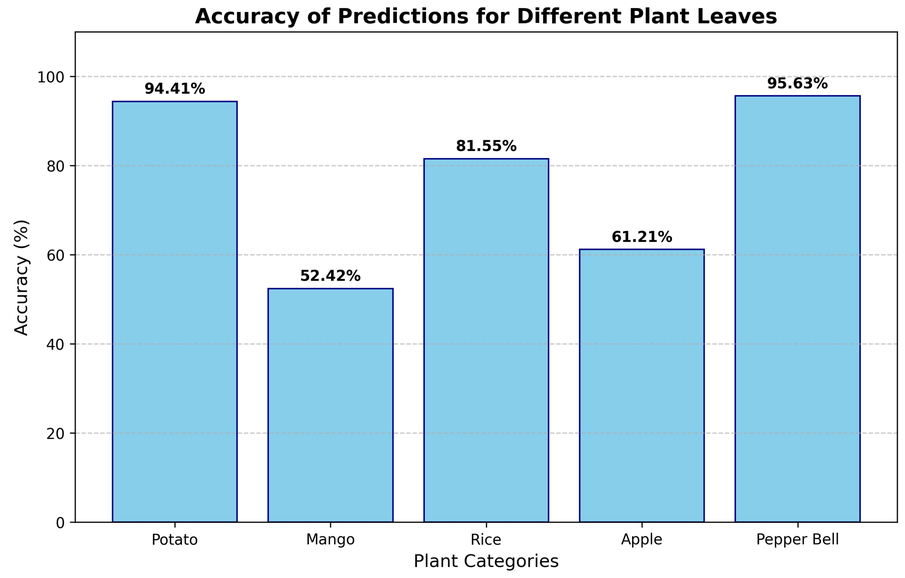

Figure 4. Accuracy comparison across plant leaf species.

Table 1. Performance comparison against existing models.

Model	Type	Acc. (%)	Key Focus
ITC-Net [1]	Multimodal	99.48	Image-text feature learning
Transfer Learning [2]	Transfer	86.29	PlantCLEF2022 benchmark
FSL + Transfer [3]	Few-shot	88.40	ImageNet feature refinement
Performer-Net [4]	Few-shot	99.12	Attention-based learning
AgirLeafNet [5]	Hybrid	92.00	Multi-crop classification
Proposed (SqNet+Proto)	**Few-shot**	**95.63**	**Ultra-lightweight (73k params)**

Main Contributions

- **Lightweight Design:** A highly optimized SqueezeNet backbone with only **73K parameters** — ideal for mobile and edge deployment.
- **Attention-Driven Features:** Channel and Spatial Attention mechanisms focus the model on discriminative lesion regions for more precise classification.
- **Few-Shot Efficiency:** Achieves up to **95.63% accuracy** with minimal training samples per class.
- **Scalable:** Generalizes effectively across multiple plant species (Potato, Rice, Pepper Bell, Apple, Mango).

Future Work

- **Edge Deployment:** Integrating the model into mobile applications to enable real-time disease monitoring directly in the field.
- **Architecture Enhancement:** Exploring **Vision Transformer (ViT)** backbones to further improve feature representation and classification robustness.
- **Dataset Expansion:** Broadening the framework to cover a wider variety of plant species and diverse environmental conditions such as varying lighting and weather.

Conclusion

This work presents a lightweight few-shot learning framework for plant leaf disease classification that achieves up to **95.63% accuracy** using only **73K parameters**. By combining a compact SqueezeNet backbone with a Prototypical Network and attention mechanisms, the model delivers strong performance with minimal labeled data. It offers a practical, scalable, and resource-efficient solution for real-world agricultural disease detection, particularly in data-scarce and low-resource environments.

References

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